Preface

The "Optimizing Supply Chain Efficacy using Machine Learning and Deep Learning Techniques" project investigates how modern computational approaches can improve decision-making processes and operational efficiencies in global supply chains. The project uses predictive analytics, leveraging datasets from Mendeley.com, to address key areas such as sales prediction, order quantity prediction, fraud detection, and late delivery prediction. The research aims to bridge the gap between theoretical advancements in machine learning and their practical applications in supply chain management.

The project employs a diverse array of models, including traditional machine learning techniques like Linear Regression and Decision Trees, as well as advanced deep learning methods like Long Short-Term Memory (LSTM) networks. Each model was meticulously developed and tested to ensure robust performance and high accuracy, providing actionable insights to enhance various aspects of supply chain operations.

The findings are significant for businesses seeking to optimize their supply chain strategies. Sales prediction models offer valuable foresight for inventory and procurement planning, while accurate order quantity predictions help streamline production schedules and minimize waste. Fraud detection algorithms safeguard financial transactions and late delivery prediction models enable proactive measures to ensure timely deliveries and improve customer satisfaction.

The Decision Tree model achieved impressive accuracy rates for fraud detection and late delivery prediction, while Linear Regression models demonstrated reliability in forecasting sales and order quantities. LSTM models further underscored deep learning's potential in capturing long-term dependencies and enhancing predictive performance, particularly for time-series data.